

ALARM CODE REFERENCE

IR-7000 / RC-7000 Troubleshooting Manual

Educational Sample Document.

This chapter lists common alarms. Codes are samples for academic use only.

A-120 Encoder backup battery low

Meaning: Battery voltage below threshold.

Possible causes: Aged battery; poor connector contact.

Checks: Measure battery voltage; inspect BATT1 connector.

Corrective action: Replace battery using Maintenance Manual Section 4.

A-121 Encoder data lost

Meaning: Absolute encoder position data invalid after power loss.

Possible causes: Battery removed too long; battery failure; encoder fault.

Checks: Confirm battery replaced; check encoder cables.

Corrective action: Remaster affected axes; if recurring, replace encoder.

A-210 Invalid operating mode transition

Meaning: Requested mode change is not allowed from the current state.

Possible causes: Safety gate open in Auto request; E-stop active; program running.

Checks: Release E-stop; stop program; close gates; select Teach first if required.

A-305 Controller overtemperature

Meaning: Cabinet temperature exceeded warning threshold.

Possible causes: Dirty filter; failed fan; high ambient temperature.

Checks: See Maintenance Manual Cooling System section.

SRVO-001 Motor power not ready

Meaning: Servo power circuit did not close.

Possible causes: E-stop chain open; contactor K1 failure; 24V control supply fault.

Checks: E-stop circuit continuity; K1 coil voltage; 24V PSU output.

SRVO-021 Servo overload

Meaning: RMS current exceeded thermal model limit for an axis motor.

Possible causes: Excessive payload; mechanical binding; aggressive acceleration.

Checks: Verify tool mass · rated payload; jog axis slowly; inspect for collisions.

SRVO-045 Position deviation excess

Meaning: Tracking error between command and feedback exceeded limit.

Possible causes: Collision; brake not released; encoder fault; gain mismatch.

Checks: Clear obstacles; confirm brake release; inspect encoder cable.

COMM-010 Controller communication timeout

Meaning: Pendant or fieldbus communication lost for longer than watchdog time.

Possible causes: Pendant cable damage; EMI; incorrect network configuration.

Checks: Reseat pendant cable; inspect for cuts; verify Ethernet settings.

E-123 Analog input out of range

Meaning: Configured analog input channel exceeded high/low limits.

Possible causes: Sensor failure; wiring open/short; incorrect scaling.

Checks: Measure sensor signal; verify wiring to AI channel; confirm software limits.

SERVO NOT TURNING ON

Symptom: Robot servo will not energize after MOTOR ON.

Diagnostic sequence:

1. Confirm no active E-stop on controller, pendant, or external gate.
2. Check that alarm history does not show SRVO-001 or power faults.
3. Verify 24V control supply is within 22-26 VDC.
4. In Teach mode, hold enabling device in middle position while pressing MOTOR ON.
5. Listen for contactor K1 engagement.
6. If K1 does not engage, inspect safety relay board and E-stop chain.
7. If K1 engages but axes are limp, check brake release 24V at motor connectors.

Do not defeat safety relays. If the cause is unclear, escalate to a qualified service technician and consult the official manufacturer documentation.

OVERHEATING TROUBLESHOOTING

Symptom: Robot or controller overheats during production.

If controller cabinet overheats (A-305):

- Clean filters and verify fans.
- Reduce ambient temperature.
- Check duty cycle and amplifier loading.

If a motor overheats:

- Reduce payload or cycle rate.
- Inspect for mechanical binding and brake drag.
- Confirm correct motor and amplifier pairing after any repair.

Recalibration is not a remedy for overheating. Address thermal and mechanical root causes first.

COMMUNICATION FAULTS

COMM-010 and related communication faults.

Pendant communication:

- Replace pendant cable with a known good cable.
- Keep pendant cable away from welding power leads.

Fieldbus / Ethernet:

- Verify IP address, subnet, and PLC connection.
- Check that switch ports are set to auto-negotiation unless fixed speed is required.
- Excessive packet loss may indicate EMI or a failing NIC.

After correcting wiring, reset the alarm and retry Auto operation.